

1 Gaussian Posterior Derivation

1.1 Input and Assumption Nodes

Input nodes: `likelihood.gaussian` and `prior.gaussian_beta`.

1.2 Derivation

Combining likelihood and prior gives the exponent

$$-\frac{1}{2\sigma^2}\|y - X\beta\|_2^2 - \frac{1}{2\tau^2}\|\beta\|_2^2.$$

Completing the square yields

$$\Sigma_n = \left(\frac{1}{\sigma^2} X^\top X + \frac{1}{\tau^2} I_d \right)^{-1}, \quad \mu_n = \Sigma_n \frac{1}{\sigma^2} X^\top y.$$

1.3 Result

The posterior node `posterior.gaussian_conjugate` returns $p(\beta \mid X, y) = \mathcal{N}(\mu_n, \Sigma_n)$.